

The Cost of Standard and Complex Pars Plana Vitrectomy for Retinal Detachment Repair Exceeds Its Reimbursement

Warren W. Pan, MD, PhD,¹ David S. Portney, MD,¹ Shahzad I. Mian, MD,¹ Rajesh C. Rao, MD^{1,2,3,4,5,6,7,8}

Objective: To measure the total costs and reimbursements associated with standard and complex pars plana vitrectomy using time-driven activity-based costing (TDABC).

Design: Economic analysis at a single academic institution.

Subjects: Patients who underwent standard or complex pars plana vitrectomy (PPV; Current Procedural Terminology codes 67108 and 67113) at the University of Michigan in the calendar year 2021.

Methods: Process flow mapping for standard and complex PPVs was used to determine the operative components. The internal anesthesia record system was used to calculate time estimates, and financial calculations were constructed from published literature and internal sources. A TDABC analysis was used to determine the costs of standard and complex PPVs. Average reimbursement was based on Medicare rates.

Main Outcome Measures: The primary outcomes were the total costs for standard and complex PPVs and the resulting net margin at current Medicare reimbursement levels. The secondary outcomes were the differential in surgical times, costs, and margin for standard and complex PPV.

Results: Over the 2021 calendar year, a total of 270 standard and 142 complex PPVs were included in the analysis. Complex PPVs were associated with significantly increased anesthesia time (52.28 minutes; $P < 0.001$), operating room time (51.28 minutes; $P < 0.0001$), surgery time (43.64 minutes; $P < 0.0001$), and postoperative time (25.95 minutes; $P < 0.0001$). The total day-of-surgery costs were \$5154.59 and \$7852.38 for standard and complex PPVs, respectively. Postoperative visits incurred an additional cost of \$327.84 and \$353.86 for standard and complex PPV, respectively. The institution-specific facility payments were \$4505.50 and \$4935.14 for standard and complex PPV, respectively. Standard PPV yielded a net negative margin of $-\$976.93$, whereas complex PPV yielded a net negative margin of $-\$3271.10$.

Conclusions: This analysis demonstrated that Medicare reimbursement is inadequate in covering the costs of PPV for retinal detachment, with a particularly large negative margin for more complex cases. These findings demonstrate that additional steps may be necessary to mitigate adverse economic incentives so that patients continue to have timely access to care to achieve optimal visual outcomes after retinal detachment.

Financial Disclosure(s): The authors have no proprietary or commercial interest in any materials discussed in this article. *Ophthalmology Retina* 2023;7:948-953 © 2023 by the American Academy of Ophthalmology



Supplemental material available at www.ophtalmologyretina.org.

At the end of 2022, Congress passed an updated Medicare payment model that averted the feared, planned 8.5% cut for ophthalmology, yet still resulted in a 1.5% cut for ophthalmologists.¹ Although there are efforts to address this,² when accounting for medical care and practice expense inflation, physician payments have decreased by 26% since 2001.³ These realities point to an important intersection between ophthalmic care, financial reimbursements, and health care delivery costs.

In 2021, Berkowitz et al⁴ found that the cost associated with elective pars plana vitrectomy (PPV) surgery significantly exceeded the Medicare reimbursement using time-driven activity-based costing (TDABC), a bottom-up costing methodology outlined by Kaplan and Porter.⁵

These same economic realities likely apply to nonelective PPV. As patients will continue to have retinal detachments requiring timely surgical intervention,⁶ and with rates of rhegmatogenous and tractional retinal detachments only increasing,^{7,8} an improved understanding of the costs of these surgeries is valuable.

Given the complex economic factors at play, and their potential impact on practice patterns, we sought to assess the costs associated with standard and complex retinal detachment repair and compare them with Medicare reimbursement. The comparison of cost and reimbursement allows for an understanding of the practical economic impact, which may both explain practice patterns and identify opportunities for reimbursement adjustments.

Methods

The study was exempted by the University of Michigan institutional review board and complied with the Declaration of Helsinki. Deidentified electronic health record (EHR) data (Epic Systems) were accessed for standard PPV surgery Current Procedural Terminology (CPT) code 67108 and complex PPV surgery CPT code 67113 for the calendar year 2021. Exclusion criteria were for joint operations where other ophthalmic surgeons and procedures were performed separate from the PPV. Cost analysis was performed using the TDABC approach.^{4,5,9}

Process Flow

A flow diagram was created based on discussions and case observations. The processes included preoperative, perioperative, and postoperative periods determined by the timestamps of each case as extracted from the EHR. The costs associated with each arena included the following: (1) preoperative nursing care; (2) perioperative surgeon, anesthesia, operating room (OR) time, OR personnel, supplies; and (3) postoperative nursing care (Fig 1). The TDABC analysis was limited to the day of surgery given the inability to accurately quantify the costs of preoperative and postoperative visits at the institution of interest. The number of postoperative visits in the global billing period was separately calculated, and a separate, more rudimentary analysis was used to estimate their costs.

Time Estimates

Time data were acquired from the University of Michigan's EHR and via DataDirect (Precision Health). Surgery-related data were downloaded to Microsoft Excel (Microsoft) and included the CPT billing code, date, surgeon, and time stamps for all steps associated with the event, including preoperative assignment, OR start, anesthesia start, surgery start, surgery end, anesthesia end, OR end, and postoperative end times. Any incomplete timestamp data were manually examined and filled from the EHR.

Time measurements were calculated similarly to a previously published study (Supplemental Methods, available at www.opthalmologyretina.org).⁹ Preoperative minutes were calculated as the difference between preoperative assignment time and OR start time. Anesthesia minutes were calculated as the difference between anesthesia start and anesthesia end times. Operating room minutes were calculated as the difference between OR start and OR end times. Surgery minutes were calculated as the difference between surgery start and surgery end times and excluded preparation and draping time.

Postoperative minutes were calculated as the difference between anesthesia end and postoperative end times.

Labor and Space Costs

Our analysis of PPVs was restricted to a single academic health center, where surgeons and personnel are employed and the OR facilities and instruments owned by the same entity. Capacity cost rates were calculated in a manner similar to a previously published study (Supplemental Methods, available at www.opthalmologyretina.org).⁹ Surgeon costs that were estimated from the department incorporated salary, bonuses, benefits, and amount of time worked. Of note, we were unable to specifically identify retina surgeon costs because of concerns for confidentiality. Anesthesia costs were calculated with a top-down approach using total allocated anesthesia costs and total OR minutes in 15-minute increments. Associated OR costs, including personnel (e.g., circulating nurses, scrub technicians), were estimated based on a previous study and adjusted for inflation.¹⁰ Similarly, preoperative and postoperative costs were estimated from a previous study and adjusted for inflation using the United States government Consumer Price Index data.¹¹

Supplies Costs

The EHR contains a list of all supplies and tools used for each surgery, along with the supply costs. The total supply costs were included with each encounter. Additionally, the use of laser, cryoprobe, tamponade agent, scleral buckle, membrane peel, staining agent (e.g., triamcinolone, Brilliant blue), perfluorocarbon liquid, scleral buckle, and retinectomy were specifically documented for each case. Importantly, medication costs outside of those agents used during the surgery were not documented in the EHR and were therefore excluded from the analysis.

Postoperative Clinic Visits

The number of postoperative visits within 90 days after vitrectomy was counted to reflect the global period. Only visits to the retina clinic and for postoperative care qualified. We applied a previously published estimate of cost of \$86.73 per postoperative clinic visit to assess the incremental costs of these visits,¹² assuming that time and effort involvement are comparable for standard and complex PPVs.

Reimbursement Rates

The average facility payments in the United States were identified based on the CPT codes 67108 and 67113. Medicare reimbursement rates were extracted from the Centers for Medicare and

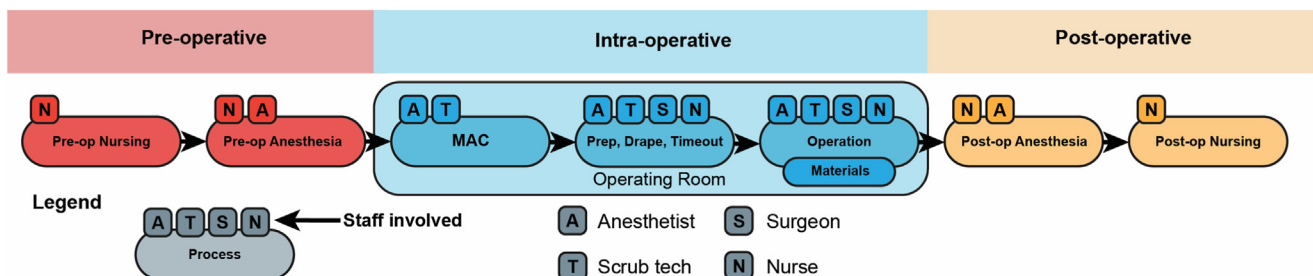


Figure 1. Flow diagram process for standard and complex vitrectomies divided into preoperative, intraoperative, and postoperative periods. MAC = monitored anesthesia care; Post-op = postoperative; Pre-op = preoperative.

Table 1. Time and Cost Contributions to, and Comparison between, Standard and Complex Pars Plana Vitrectomy Surgeries*

	Time (min)	Time Difference (min)	Costs (\$)	Cost Difference (\$)
Preoperative	Standard (67 108) 89.96	7.01 (0.20)	Standard (67 108) 62.97	-4.91
	Complex (67 113) 82.04		Complex (67 113) 58.06	
Anesthesia	Standard (67 108) 106.66	52.28 (< 0.001)	Standard (67 108) 316.78	155.27
	Complex (67 113) 158.94		Complex (67 113) 472.05	
OR	Standard (67 108) 93.43	51.28 (< 0.001)	Standard (67 108) 3492.41	1916.85
	Complex (67 113) 144.71		Complex (67 113) 5409.26	
Surgery	Standard (67 108) 71.95	43.64 (< 0.001)	Standard (67 108) 259.74	157.54
	Complex (67 113) 115.59		Complex (67 113) 417.28	
Postoperative	Standard (67 108) 61.54	25.95 (< 0.001)	Standard (67 108) 43.08	18.17
	Complex (67 113) 87.49		Complex (67 113) 61.24	
Time totals			Standard (67 108) 4174.98	2242.92 (< 0.001)
			Complex (67 113) 6417.89	
Materials			Standard (67 108) 979.60	454.89 (< 0.001)
			Complex (67 113) 1434.49	
Grand total			Standard (67 108) 5154.59	2697.80 (< 0.001)
			Complex (67 113) 7852.38	

OR = operating room.

*Surgeon time costs were estimated from our institution's ophthalmologist salaries and are not specific to retinal surgeons because of institutional concerns regarding salary disclosure from a relatively small number of surgeons. Therefore, these costs are likely an underestimate of the true surgeon time costs.

Medicaid Services (CMS) for physician fees specific for Michigan.¹³ Anesthesia reimbursements were calculated from CPT code 00145 with Michigan conversion and calculated anesthesia time.¹³

National average facility payments to hospital outpatient departments were identified by the CPT codes 67108 and 67113.¹⁴ Additionally, our institution-specific facility payments were queried and obtained through the University of Michigan.

Analysis

Microsoft Excel was used to determine the associated costs to determine the difference in costs and surgery time between standard (CPT 67108) and complex (CPT 67113) PPV surgeries. GraphPad Prism (GraphPad) was used for statistical analysis.

Results

There were 270 standard vitrectomies (CPT 67108) and 142 complex vitrectomies (CPT 67113) performed after excluding 9 vitrectomies (1 standard and 8 complex), which were combined with another ophthalmic surgery in the 2021 calendar year. Of the 270 standard vitrectomies and 142 complex vitrectomies, 8 and 31

involved the use of scleral buckles, respectively. The average times for standard PPV were divided into preoperative (89.95 minutes), anesthesia (106.66 minutes), OR (93.43 minutes), surgery (71.95 minutes), and postoperative times (61.54 minutes) (Table 1, Fig 2). There were significant differences when complex PPV anesthesia (158.94 minutes; $P < 0.001$), OR (144.71 minutes; $P < 0.0001$), surgery (115.59 minutes; $P < 0.001$), and postoperative times were compared (87.49 minutes; $P < 0.001$) but not when the preoperative time was compared (82.04 minutes; $P = 0.2$) (Table 1, Fig 2).

The day-of-surgery time-associated resource costs totaled \$4174.98 and \$6417.89 for standard and complex PPVs, respectively (Table 1, Fig 3). Operating room costs accounted for 83.7% and 84.3% of time-based costs for standard and complex PPV, respectively. Surgeon costs accounted for 6.2% and 6.5% of standard and complex costs, respectively. Supplies costs added \$979.60 and \$1434.39 for standard and complex PPV, respectively. The total day-of-surgery costs for standard and complex vitrectomy were \$5154.59 and \$7852.38, respectively (Table 1, Fig 3). Over 83% of the day-of-surgery cost differential between complex and standard vitrectomies was attributed to the difference in time, whereas the supplies costs accounted for around 16% (Fig 3).

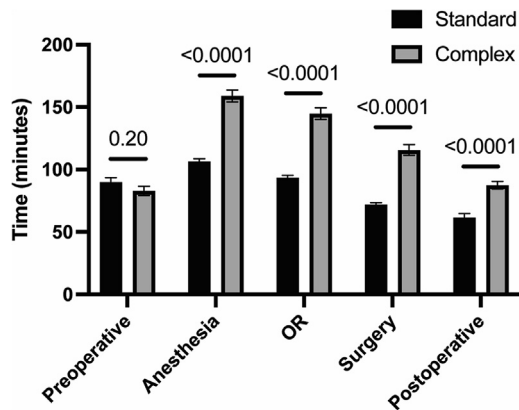


Figure 2. Average minutes spent in the preoperative, anesthesia, operating room, surgery, and postoperative stages associated with standard (black bars) and complex (gray bars) vitrectomies. OR = operating room.

Patients who underwent complex PPV had a higher mean number of postoperative clinic visits within the global period (4.08) than those who underwent standard vitrectomies (mean, 3.78; $P = 0.036$) (Fig 4). With the aforementioned cost of \$86.73 per postoperative clinic visit,¹² standard and complex PPVs incur an additional cost of \$327.84 and \$353.86, respectively. After accounting for postoperative visits, the total costs of standard and complex PPVs increase to \$5482.43 and \$8206.24, respectively.

The CMS professional fee reimbursement for the CPT codes 67108 and 67113 are \$1,224.34 and \$1,369.27, respectively, in Michigan.¹³ For anesthesia (CPT code 00145 calculated with Michigan conversion factor and calculated anesthesia time), the reimbursement on average was \$293.29 and \$371.25 for standard and complex vitrectomies, respectively. The institution-specific facility payments were \$2987.87 and \$3194.62 for standard and complex vitrectomies, which equates to a total reimbursement of \$4505.50 and \$4935.14, respectively.

In total, the net margin of standard PPV was -\$976.93, whereas that of complex PPV was -\$3271.10 for each surgery performed. This calculation does not encompass the costs associated with OR turnover time, which would likely increase the costs and further reduce the margin for these operations.

Discussion

In this study, we utilized TDABC to quantify the surgical costs of standard and complex vitrectomy for retinal detachment and compared these costs with reimbursement rates. Our TDABC model detailed a total cost of \$5482.43 and \$8206.24 for standard and complex PPV surgery, respectively. The institution-specific reimbursements for standard and complex PPV surgery were \$4505.50 and \$4935.14, respectively. Given the associated costs for these surgeries, the aforementioned reimbursement rates are inadequate in covering the costs associated with both standard and complex vitrectomy surgeries. Indeed, 82% of standard PPV costs are covered by average reimbursement, whereas the figure is only 60% for complex PPV. This

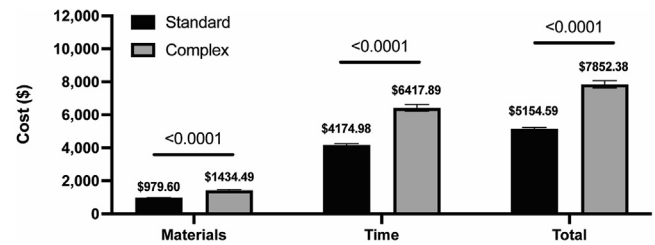


Figure 3. Average supplies, time, and total costs associated with standard (black bars) and complex vitrectomies (gray bars).

accounts for a large negative margin for PPV for retinal detachment, particularly for complex PPV.

Further analysis of the important drivers of surgical costs identified OR time as the predominant factor for both standard and complex vitrectomy. The OR time alone accounts for 110% of reimbursement for complex vitrectomy. Importantly, our analysis of costs did not include OR turnover time and, therefore, underestimates the total costs associated with these surgeries and the associated negative margin for performing them. Furthermore, it is worth noting that CMS does not provide variable reimbursement for the actual costs of supplies (although some insurance plans do reimburse for the supplies individually) and instead, the flat-rate facility payment is meant to cover these costs. Such distinction is necessary because not only is the provider required to absorb the time-based costs of surgeries but they must also absorb the supply expense.

It is important to evaluate the findings of our analysis in the context of the practice environment for which it was conducted. The use of a single academic institution in our study is a notable limitation as it may prevent the extrapolation of our findings to other retina practices nationally. Still, this same limitation allows for a more precise analysis. Our calculated complex PPV costs for retinal detachments are comparable with the costs previously calculated by Berkowitz et al,⁴ whereas standard PPV for retinal detachments costs significantly less. Cost inputs, including OR costs, in addition to supply expenses, may explain the relative differences in costs. Moreover, the exclusion of costs such as OR turnover time from our

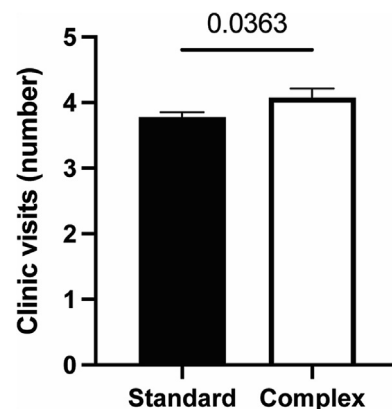


Figure 4. Average number of postoperative clinic visits for standard (black bar) or complex vitrectomies (white bar) over 90 days after surgery.

study, which was estimated to cost \$1030.39 by Berkowitz et al,⁴ likely explains much of the difference noted. Not only will costs differ between institutions but also reimbursement. By way of comparison, the national average facility payment established by CMS for hospital outpatient departments for these CPT codes is \$3999, which coincides with a total average payment of \$5223.34 and \$5368.27 for standard and complex vitrectomies, respectively.¹⁴

Although important limitations exist to this study, the findings remain crucial. A key limitation of this study is the restriction of this cost analysis to 1 academic institution, as such costs and reimbursements may vary from practice to practice. Analysis of privately owned practices may be quantitatively different on both the cost and reimbursement sides. Therefore, further investigations breaking down the individual components that may affect margins with differing cost structures would further contribute to our understanding of the landscape. Furthermore, we applied broad assumptions to post-operative clinical care costs, which may be variable among institutions¹⁵ and may muddle the day-of-surgery cost analyses. Despite these limitations, our analyses clearly demonstrate that CMS reimbursements are inadequate in covering the costs of PPV for retinal detachments, especially complex PPV.

Concerns for limited reimbursement for retinal surgery may encourage increased utilization of nonsurgical

interventions or an increased allocation of time to clinic.¹² In a recent article by the American Academy of Ophthalmology,¹⁶ Leung et al¹⁷ highlighted the significant opportunity cost for vitreoretinal surgeons to spend time in surgery compared with that in the clinic. A recent perspective in the *American Journal of Ophthalmology* echoes this by highlighting the financial pressures with the lucrative path of clinical visits and intravitreal anti-VEGF injections when compared with PPV.¹² Here, we further demonstrate the financial pressures and negative margin associated with PPV.

In a world where both rhegmatogenous and tractional retinal detachments have increased significantly over the last 2 decades,⁸ as well as the rising rates of high myopia¹⁸ and diabetes,¹⁹ we likely face a future with more numerous retinal detachments requiring more complex surgeries. With patient visual outcomes tied to time of intervention,⁶ there is a need to minimize adverse economic incentives in order to ensure patients achieve timely access to both standard and complex vitreoretinal surgical care. One solution, as Leung et al¹⁷ noted, may be to increase reimbursement for complex vitrectomies. Fortunately, there is a precedent for this with the increase in reimbursement rates for cataracts-minimally invasive glaucoma surgery codes.²⁰ Therefore, there exists a clear and present need for Medicare reimbursements for standard and complex vitrectomies to more closely match the cost realities.

Footnotes and Disclosures

Originally received: March 31, 2023.

Final revision: June 21, 2023.

Accepted: June 26, 2023.

Available online: July 1, 2023. Manuscript no. ORET-D-23-00019R2.

¹ Department of Ophthalmology and Visual Sciences, Kellogg Eye Center, University of Michigan, Ann Arbor, Michigan.

² Department of Pathology, University of Michigan, Ann Arbor, Michigan.

³ Department of Human Genetics, University of Michigan, Ann Arbor, Michigan.

⁴ Rogel Cancer Center, University of Michigan, Ann Arbor, Michigan.

⁵ Center of Computational Medicine and Bioinformatics, University of Michigan, Ann Arbor, Michigan.

⁶ Center for RNA Biomedicine, University of Michigan, Ann Arbor, Michigan.

⁷ A. Alfred Taubman Medical Research Institute, University of Michigan, Ann Arbor, Michigan.

⁸ Section of Ophthalmology, Surgery Service, Veterans Administration Ann Arbor Healthcare System, Ann Arbor, Michigan.

Disclosures:

All authors have completed and submitted the ICMJE disclosures form.

The authors have no proprietary or commercial interest in any materials discussed in this article.

R.C.R.: Supported — National Eye Institute (R01EY030989, P30EY007003), the National Cancer Institute (P30CA046592, to the University of Michigan Rogel Comprehensive Cancer Center), and Research to Prevent Blindness (Departmental Grant to the University of Michigan Kellogg Eye Center and Career Advancement Award), Beatrice and Reymont Paul Foundation, March Hoops to Beat Blindness, the Taubman

Institute, the Leonard G. Miller Endowed Professorship and Ophthalmic Research Fund at the Kellogg Eye Center, and the Grossman, Elaine Sandman, Marek and Maria Spatz (endowed fund), Greenspon, Dunn, Avers, Boustikakis, Sweiden, and Terauchi research funds.

HUMAN SUBJECTS: No human subjects were included in this study. The study was exempted by the University of Michigan institutional review board and complied with the tenets of the Declaration of Helsinki. This is a retrospective study using de-identified subject details. Informed consent was not obtained.

No animal subjects were included in this study.

Author Contributions:

Conception and design: Pan, Portney, Mian, Rao

Data collection: Pan, Portney, Mian, Rao

Analysis and interpretation: Pan, Portney, Mian, Rao

Obtained funding: NA

Overall responsibility: Pan, Portney, Mian, Rao

Abbreviations and acronyms:

CPT = Current Procedural Terminology; **EHR** = electronic health record; **OR** = operating room; **PPV** = pars plana vitrectomy; **TDABC** = time-driven activity-based costing.

Keywords:

Retina, Pars plana vitrectomy, Time-driven activity-based costing, Reimbursement, Health Policy.

Correspondence:

Rajesh C. Rao, MD, University of Michigan Medical School, W.K. Kellogg Eye Center, 1000 Wall St., Ann Arbor, MI 48105. E-mail: rajeshr@umich.edu.

References

1. American Academy of Ophthalmology. Congress passes year-end deal that will mitigate 2023 Medicare payment cuts. <https://www.aao.org/eye-on-advocacy-article/congress-year-end-deal-mitigate-2023-medicare#:~:text=Instead%20of%20stopping%20the%20entire,cut%20to%202%25%20in%202023>. Accessed April 30, 2023.
2. Library of Congress. H.R.2474 - 118th Congress (2023-2024): strengthening Medicare for Patients and Providers Act. <https://www.congress.gov/bill/118th-congress/house-bill/2474>. Accessed April 14, 2023.
3. American Medical Association. Medicare physician payment is not keeping up with inflation. <https://www.ama-assn.org/system/files/medicare-updates-inflation-chart.pdf>. Accessed April 30, 2023.
4. Berkowitz ST, Sternberg P, Patel S. Cost analysis of routine vitrectomy surgery. *Ophthalmol Retina*. 2021;5:496–502.
5. Kaplan RS, Porter ME. How to solve the cost crisis in health care. *Harv Bus Rev*. 2011;89:46–52.
6. Mahmoudi S, Almony A. Macula-sparing rhegmatogenous retinal detachment: is emergent surgery necessary? *J Ophthalmic Vis Res*. 2016;11:100–107.
7. Reeves MG, Pershing S, Afshar AR. Choice of primary rhegmatogenous retinal detachment repair method in US commercially insured and Medicare advantage patients, 2003–2016. *Am J Ophthalmol*. 2018;196:82–90.
8. Madi HA, Keller J. Increasing frequency of hospital admissions for retinal detachment and vitreo-retinal surgery in England 2000–2018. *Eye*. 2022;36:1610–1614.
9. Portney DS, Berkowitz ST, Garner DC, et al. Comparison of incremental costs and Medicare reimbursement for simple vs complex cataract surgery using time-driven activity-based costing. *JAMA Ophthalmol*. 2023;141:358–364.
10. Childers CP, Maggard-Gibbons M. Understanding costs of care in the operating room. *JAMA Surgery*. 2018;153:e176233.
11. Yangyang RY, Abbas PI, Smith CM, et al. Time-driven activity-based costing to identify opportunities for cost reduction in pediatric appendectomy. *J Pediatr Surg*. 2016;51:1962–1966.
12. Young BK, Hwang M, Johnson MW, et al. A caveat about financial incentives for anti-vascular endothelial growth factor therapy for diabetic retinopathy. *Am J Ophthalmol*. 2022;243:77–82.
13. Centers for Medicare and Medicaid Services. Physician fee schedule. <https://www.cms.gov/medicare/physician-fee-schedule/search>. Accessed January 9, 2022.
14. Medicare.gov. Repair of retinal detachment. <https://www.medicare.gov/procedure-price-lookup/cost/67108/>. Accessed January 9, 2022.
15. Imran JB, Madni TD, Taveras LR, et al. Analysis of operating room efficiency between a hospital-owned ambulatory surgical center and hospital outpatient department. *Am J Surg*. 2019;218:809–812.
16. Patel S, Leung E. Opportunity costs of vitreoretinal surgeries. <https://www.aao.org/interview/opportunity-costs-of-vitreoretinal-surgeries>. Accessed November 11, 2022.
17. Leung EH, Leder HA, Patel S, et al. Opportunity cost of retinal detachment surgery vs office-based patient care. *J VitreoRetinal Dis*. 2022;6:278–283.
18. Holden BA, Fricke TR, Wilson DA, et al. Global prevalence of myopia and high myopia and temporal trends from 2000 through 2050. *Ophthalmology*. 2016;123:1036–1042.
19. Saeedi P, Petersohn I, Salpea P, et al. Global and regional diabetes prevalence estimates for 2019 and projections for 2030 and 2045: results from the International Diabetes Federation Diabetes Atlas. *Diabetes Res Clin Pract*. 2019;157:107843.
20. AAO. Top four issues threatening ophthalmology's financial stability. <https://eyewire.news/news/aa-top-four-issues-threatening-ophthalmologys-financial-stability?c4src=article:infinite-scroll>. Accessed November 12, 2022.